

Please describe three examples of the most important variables needed to address the key question about how aspects of places where people lived, as well as their individual characteristics and behaviours, influenced risk of COVID 19 infection during the recent pandemic? Why are these variables important?

To address this question, it is crucial to first understand how COVID-19 infects people. The most direct and robust evidence comes from the human SARS-CoV-2 challenge study (Zhou et al., 2023). First, the primary mode of transmission is via airborne aerosols, although direct contact and indirect contact through contaminated surfaces also play a role. Thus, it is important to understand how local environmental conditions sustain these aerosols. Second, two superspreading individuals were responsible for 86% of viral emissions, highlighting the need to examine the role of superspreading events and how place-related factors may contribute to their occurrence. Third, only 7% of viral emission occurs when individuals are asymptomatic. Early testing against initial signs of infection can prevent community transmission. Therefore, it is important to explore the impact of early testing within the context of places.

Three factors can be noted by examining the causal pathways of COVID-19 infection, which include the levels of ventilation, the frequency of random contacts, and the delay in diagnosis. Each of these factors can be proxied by relevant variables for statistical modelling. Although there are some other variables, e.g. age, ethnicity, etc. (Feng, 2023), that have been found to be important, it is interesting and perhaps more meaningful to follow causal pathways (Huenermund et al., 2022).

Factor 1: The Level of Ventilation

It is often assumed that Covid-19 would display a seasonal pattern as it became endemic, influenced by humidity and temperature (Byun et al., 2021; Townsend et al., 2023). However, animal studies suggest that these factors have little effect on transmissibility (Yinda et al., 2024). Consequently, any observed seasonality is likely driven by unobserved confounders, such as changes in human behaviour under different weather conditions.

Interestingly, evidence indicates that low wind speeds in outdoor environments are associated with a higher likelihood of infection (Clouston et al., 2021; Rendana, 2020). Moreover, good ventilation has been shown to remove aerosols from indoor spaces, reducing the risk of infection. Increasing measurements of indoor CO₂ concentrations—as a proxy for air quality—have raised concerns about poor ventilation due to housing designs and overpopulation (Morawska et al., 2024). Therefore, I suspect that housing quality may be an important variable to consider.

A recent study in England employed housing tenure and the number of housing problems, as recorded in the English Longitudinal Study of Ageing, as a proxy for housing quality (Hu et al., 2025). Similar research in Scotland did not include housing problems; however, the Scottish House Condition Survey remains the only national survey to examine the physical condition of homes in Scotland (Scottish Government, 2025). This survey covers household characteristics and tenure, from which comparable data items may be derived.

Factor 2: Frequency of Random Contact

Predicting superspreading events based on individual characteristics is challenging due to the complex biological context (Nielsen et al., 2023; Sneppen et al., 2021). Analysis of contact tracing across England and Wales shows that, although family contacts account for only 6% of total interactions, they are responsible for over 40% of transmission (Ferretti et al., 2024). Nevertheless, worldwide, household infection rates typically remain below 20% (Sneppen et al., 2021). Therefore, without superspreading events occurring outside of family networks, Covid-19 transmission would be largely limited and clustered across certain families.

Agent-based modelling suggests that, in the context of superspreading, random contacts in pubs and bars, where contact is highly heterogeneous, pose the greatest risk, compared to work, school, and home settings, where people tend to engage with relatively fixed groups (Kuylen et al., 2022; Sneppen et al., 2021). Thus, before the lockdown measures, proximity to pubs may be a significant variable (Somkid et al., 2020), particularly as customers tend to prefer visiting establishments close to home (Wells et al., 2023). These location data might be obtainable from premises licence applications to local governments.

Due to the lockdown and social distancing, pubs and bars were closed. It is still intriguing to explore other points of public gatherings. A potential source of data is the contact tracing apps, such as Check In Scotland and Protect Scotland. These apps provide information about how many people have visited the same locations, which could be particularly valuable for analysing the later phases of the pandemic.

Factor 3: Delay in Diagnosis

How early people realise they have Covid-19 may depend on testing frequency and turnaround time (Larremore et al., 2021), which are associated with people's ability to access testing. An observational study in Liverpool revealed that rapid lateral flow SARS-CoV-2 antigen testing uptake was socially and spatially unequal (Green et al., 2021). Their models showed that uptake and repeat testing were lower in areas of higher deprivation, in areas located further from test sites, and among populations less confident in using Internet technologies. Positive cases were spatially clustered in deprived areas, especially those located further from test sites.

There are several potential proxies. First, social deprivation data for small areas are available from the Scottish Index of Multiple Deprivation (SIMD). Second, the location of testing sites and shipments of testing kits are available by combining data from the UK Health Security Agency and the NHS Scotland. Third, the penetration rate of contact tracing apps can be also a good proxy of early diagnosis.

Possible confounders

Individual characteristics play a crucial role in influencing outcomes related to COVID-19. Genetic predispositions, coupled with similar lifestyle patterns, often correlate with observable racial and gender trends in susceptibility and severity (Cappadona et al., 2023; SeyedAlinaghi et al., 2021). An individual's knowledge about COVID-19 is also a significant factor and can often be linked to their highest educational qualification, which shapes their understanding of health information and their subsequent behaviour (Spring, 2020). Furthermore, the presence of pre-existing health conditions, particularly in those who have experienced recent hospitalisation, can exacerbate the risks associated with COVID-19 infection (Treskova-Schwarzbach et al., 2021). This is particularly relevant to older populations (Bovil et al., 2023).

In addition to these personal factors, virological characteristics must also be considered. The virulence of the virus is largely dependent on the local SARS-CoV-2 variants circulating within a community, which can vary considerably by region (Zhao et al., 2020). Meanwhile, the level of herd immunity, influenced by the progress of local immunisation efforts, plays a vital role in limiting the spread and impact of the virus within populations (Xu et al., 2025).

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